

MEET: Towards Memory-Efficient Temporal Sparse Deep Neural Networks

Zeqi Zhu^{*†} Ibrahim Batuhan Akkaya^{*} Luc Waeijen^{*} Egor Bondarev[†]
Arash Pourtaherian^{*} Orlando Moreira^{*}
^{*}Snap Inc. [†]Eindhoven University of Technology

Abstract

Deep Neural Networks (DNNs) are accurate but compute-intensive, leading to substantial energy consumption during inference. Exploiting temporal redundancy through Δ - Σ convolution [26] in video processing has proven to greatly enhance computation efficiency. However, temporal Δ - Σ DNNs typically require substantial memory for storing neuron states to compute inter-frame differences, hindering their on-chip deployment. To mitigate this memory cost, directly compressing the states can disrupt the linearity of temporal Δ - Σ convolution, causing accumulated errors in long-term Δ - Σ processing. Thus, we propose **MEET**, an optimization framework for **ME**Memory-**EE**fficient **T**emporal Δ - Σ DNNs. MEET transfers the state compression challenge to a well-established weight compression problem by trading fewer activations for more weights and introduces a co-design of network architecture and suppression method to optimize for mixed spatial-temporal execution. Evaluations on three vision applications demonstrate a reduction of $5.1 \sim 13.3 \times$ in total memory compared to the most computation-efficient temporal DNNs, while preserving the computation efficiency and model accuracy in long-term Δ - Σ processing. MEET facilitates the deployment of temporal Δ - Σ DNNs within on-chip memory of embedded event-driven platforms, empowering low-power edge processing.

1. Introduction

Recent breakthroughs in deep learning have accelerated advancements across various computer vision tasks, prompting practitioners to increasingly adopt deep neural networks (DNNs) for performance improvement. However, this rapid progress is accompanied by significant trade-offs, as modern DNN inference requires considerable energy, hindering their deployment in resource-limited environments.

The brain-inspired computing paradigm [4, 34, 39] has garnered considerable attention as a promising solution for low-power machine intelligence. Unlike conventional von Neumann computer architectures (e.g., GPU, CPU), event-driven platforms [1, 5, 13, 20, 59, 65], inspired by the brain's

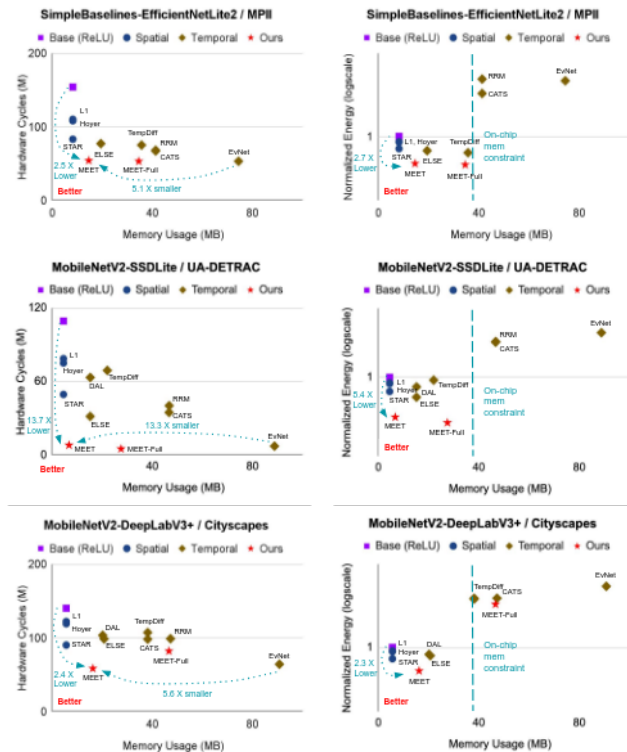


Figure 1. **MEET** inference performance in computation cycles and energy consumption across three applications. In general, temporal suppression methods outperform spatial methods in computation reduction (left); however, they may compromise energy efficiency due to memory overflow (right). MEET meets the on-chip memory constraints of an embedded event-driven platform GrAI-VIP [59], achieving optimal computation and energy savings. All evaluated models stay within 1.0% accuracy tolerance.

functional principles, bring memory closer to computing units and use dataflow architectures to exploit activation sparsity. The pivotal optimization for these platforms lies in deliberate suppression of the non-zero input activations (events) of computation-intensive layers (e.g., Conv2D), thereby resulting in diminished event-triggered (dynamic) computations. Recent research has shown that spatial suppression efficiently minimizes event count for single-frame DNN inference [21, 37, 47, 66, 71, 72], while temporal suppression, utilizing Δ - Σ modulation on linear layers [26], significantly improves the suppression performance by ex-

ploiting temporal redundancy in videos [7, 18, 49, 50, 64, 70].

Despite the superior performance of temporal suppression, their memory usage often exceeds the on-chip memory (SRAM) constraints of embedded event-driven platforms [5, 12, 59], primarily due to the memory bottleneck associated with neuron states. In particular, neuron states store the convolution outputs to yield identical results between standard convolution and temporal Δ - Σ convolution [48, 49]. Therefore, directly compressing or skipping neuron states [14, 27] can disrupt the linearity of Δ - Σ modulation, causing accumulated errors in long-term Δ - Σ processing. While storing states in off-chip DRAM could provide a solution [7, 50], it incurs approximately $100\times$ the energy cost of on-chip access [31], negating the energy savings from sparse computation reduction, as shown in Fig. 1. This poses an emerging challenge: achieving low dynamic computations through temporal suppression while ensuring that temporal Δ - Σ DNNs meet the on-chip memory constraints of embedded event-driven platforms.

To tackle this challenge, we propose an optimization framework, **MEET**, designed for **ME**emory-**EF**icient **T**emporal Δ - Σ Deep Neural Networks. MEET is developed upon three fundamental insights in event-driven processing. Initially, we observe that increasing network computations can be a viable option if it improves other critical metrics (e.g. accuracy, memory footprint), since increased computations are likely eliminated by temporal sparsity (see Fig. 2). This allows us to redesign network architectures to accommodate hardware constraints, even with increased weight count and static computations. Moreover, model accuracy scales with static computations [55]. Since each multiply-accumulate (MAC) operation embodies the inner product of an activation and a weight, we hypothesize that augmenting the weight count can reduce the activation count (neuron count) without sacrificing accuracy, and vice versa. Given the linear relationship between state size and activation count, we introduce a Compute- and State Memory-aware Network Architecture Search (CSM-NAS), which reduces states by trading fewer activations for more weights (A $\rightarrow$ W). Finally, the predominant state memory overhead arises from shallow layers, due to their large feature size. In contrast, spatial suppression [62, 69, 72] induces substantial sparsity in large features and processes frames independently, potentially mitigating the state bottleneck in shallow layers. We analyze the computation and memory costs of network architectures when incorporating different suppression methods, subsequently co-designing them to achieve a highly sparse, mixed-executed DNN with minimal states.

As illustrated in Fig. 2, at the top, MEET networks attains comparable dynamic computations and accuracy as MobileNetV2 [51] through temporal suppression [18], despite having higher static computations. At the bottom, A $\rightarrow$

W tradeoff first transfers the state compression challenge to a weight compression problem, then applies a mixed spatial-temporal suppression to alleviate the state memory bottleneck. Finally, by leveraging well-established weight compression methods [9, 38, 63, 68], MEET readily reduces the increased weights, resulting in a substantial decrease in overall memory usage for temporal Δ - Σ models.

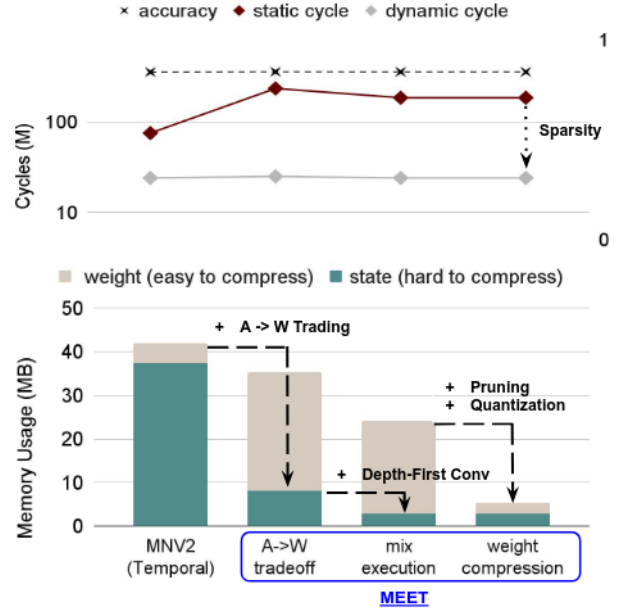


Figure 2. **Overall memory reduction in Temporal MobileNetV2 [51] without compromising accuracy and dynamic computations.** (Top) MEET leverages event sparsity to close the computation gap between light-weight and heavy-weight models. (Bottom) MEET transfers the state compression challenge to a weight compression issue, allowing for substantial memory reduction while maintaining accuracy in long-term Δ - Σ processing.

To show the effectiveness of MEET, we first establish our references by applying spatial and temporal suppression to those compute-efficient mobile architectures [51, 55]. Given that those networks are considered the state-of-the-art (SOTA) in the tradeoff between accuracy and static MAC, when integrating the activation suppression, they can yield SOTA dynamic computations. As depicted in Fig. 1, MEET not only achieves the lowest dynamic computations (i.e., hardware cycles) but also overcomes the memory bottleneck of other temporal DNNs, leveraging near-memory computing to efficiently transform computation reduction into energy savings on an event-driven platform.

Our paper presents three main contributions:

1. We propose MEET, a memory optimization framework designed to enable temporal Δ - Σ DNNs to fit within the on-chip memory constraints of event-driven platforms, empowering low-power edge processing.
2. We introduce a compute- and memory-aware NAS that trades fewer activations for increased weights, allow-

ing for the compression of neuron states without compromising accuracy in long-term Δ - Σ processing.

3. We evaluate MEET across three popular vision applications, demonstrating superior computation, memory and energy efficiency on an event-driven platform GrAI-VIP, compared to other SOTA approaches.

2. Related Work

2.1. Event-Driven Temporal Δ - Σ DNNs

Quantized Δ - Σ networks [36, 48] apply Δ - Σ modulation on linear DNN layers and quantize the input activations to minimize redundant processing between video frames. DAL [64] and CATS [70] explore learnable activation thresholds to enhance suppression, while pushing activations to low bits risks significant accuracy drop. RRM [49] and CBInfer [7] address this by truncating only small values in Δ ; however, this disrupts the linearity of Δ - Σ modulation, causing error accumulation in long-term processing. Recently, EvNet [18] and DeltaCNN [50] preserve accuracy by encoding delta truncations as states within Δ - Σ modulation. Notably, applying Δ - Σ modulation at each layer demands substantial states, and encoding delta truncations even doubles the size. Prior studies often underestimate the state size of temporal DNNs, exceeding on-chip memory constraints and resulting in high energy consumption.

2.2. Memory-Efficient Event-Driven Processing

To reduce memory overhead, previous works [14, 27] compute the input Δ while deferring Σ accumulation after a few convolution blocks, reducing intermediate neuron states. However, these methods are ineffective for nonlinear layers (e.g., ReLU) and lead to error accumulation in long-term Δ - Σ processing. DAL [64] selectively applies temporal execution based on the memory budget to maintain accuracy, but this can result in significant temporal sparsity loss if unselected layers are computationally intensive. An alternative research direction is spatial suppression [62, 69, 72], which induces activation sparsity using a single-frame input, thereby eliminating temporal dependencies and rendering the model stateless. ELSE [69] combines selective temporal execution with spatial suppression on unselected layers to offset temporal sparsity loss. However, arbitrary combinations of network architectures and suppression methods can lead to suboptimal results. Distinct from those methods, MEET addresses the state memory issue in temporal Δ - Σ DNNs by reducing activations and co-designing network and suppression method, while preserving computation efficiency and error-free long-term Δ - Σ processing.

3. Motivation

Brain-inspired event-driven platforms adopt a non-von Neumann computer architecture that leverages activation spar-

sity with near-memory computing. On these platforms, computations and memory access are triggered only by events, indicating that reducing event-triggered MACs results in proportional savings in active energy, as exhibited in Fig. 3. However, since energy efficiency varies across layer types [33, 45], MACs are not a reliable indicator of energy efficiency across different network architectures. In contrast, hardware dynamic cycles show a linear relationship with energy among all models, eliminating network architecture related noise. Thus, we use the performance estimator (GrAIFlow) provided by NeuronFlow [46], one of the efficient event-driven architectures, and adopt dynamic cycles as a key metric to compare the energy efficiency of sparse models across network architectures.

Previous studies have demonstrated that temporal suppression is the most efficient approach for reducing dynamic computations [18, 50, 64, 70]. However, they often underestimate the area and on-chip memory constraints of real-world event-driven platforms (see Fig. 4), preventing temporal DNNs from fitting entirely within SRAM. This results in costly off-chip DRAM access, negating the energy savings achieved through temporal computation reduction.

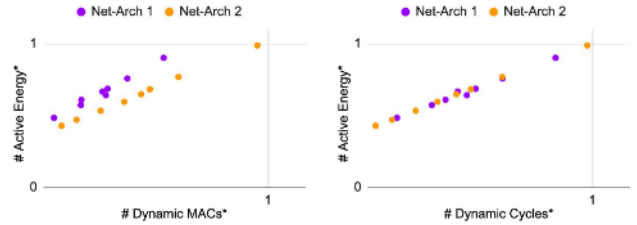


Figure 3. Normalized dynamic MACs (left) and cycles (right) vs. normalized inference energy across network architectures.

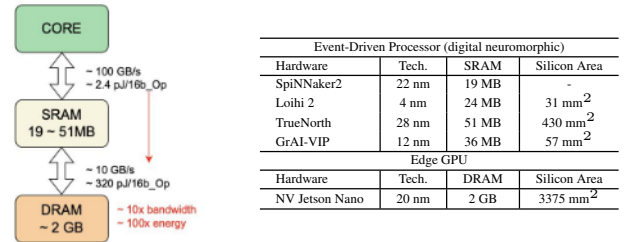


Figure 4. On-chip memory size of various event-driven platforms.

In general, the memory usage of temporal Δ - Σ DNNs can be mainly categorized into two parts: neuron states and weights. The size of neuron states varies across different temporal methods. For instance, in standard Δ - Σ DNNs [7, 36, 48, 49], the states are defined as the convolution outputs $\mathcal{A}_{state}$. In contrast, more advanced methods [18, 50] incorporate both the convolution outputs $\mathcal{A}_{state}$ and the delta-truncated outputs $\mathcal{T}_{state}$ (long-term memory) as neuron states, doubling the state size, as shown in Fig. 5. These methods achieve optimal suppression while maintaining accuracy in long-term processing. Although the state size varies across methods, it is approximately pro-

portional to the activation count (i.e., neuron count). Thus, reducing the activation count can lead to substantial state savings in temporal DNNs. However, the compression or omission of states poses a significant challenge, as it introduces errors in Δ - Σ modulation, which accumulate over the long-term Δ - Σ processing [14]. In contrast, compressing weights is a more straightforward task, as extensive research has shown that integrating sparsity with quantization can dramatically reduce the weight footprint [9, 29, 38]. Thus, we propose to trade fewer activations for increased weights in temporal Δ - Σ DNNs to reduce state size, followed by weight compression for overall memory reduction.

4. MEET

MEET comprises modular components, such as $A \rightarrow W$ tradeoff, spatial suppression, temporal suppression, weight compression, and mixed spatial-temporal execution, configurable for diverse application scenarios. MEET-Full, a fully temporal DNN, applies an $A \rightarrow W$ tradeoff along with temporal suppression and weight compression to minimize dynamic computations and memory usage. MEET-Mix@ $\lambda \times$ (λ is feature downscale factor, see Appendix C), a mixed spatial-temporal DNN, combines spatial suppression with mobile architectures in the initial stages, adopting MEET-Full’s methodology in the deeper stages. To the best of our knowledge, we are the first to address the memory challenge of temporal DNNs without compromising computation efficiency and model accuracy for long-term Δ - Σ processing.

4.1. Preliminary

Considering a DNN as a stack of $\mathcal{I}$ convolution blocks, each includes a convolution and an activation. Given an input at time t $x_t \in \mathbb{R}^{c_i \times k_h \times k_w}$ and a convolution function $\mathcal{G}$ with a kernel $w \in \mathbb{R}^{c_o \times c_i \times k_h \times k_w}$, the output $z_t \in \mathbb{R}^{c_o \times k_h \times k_w}$ is calculated as $z_t = \mathcal{G}(x_t) = w * x_t$. Furthermore, given an activation function $\mathcal{F}$ (e.g., ReLU [22]), the output $z_{t,i}$ at the i^{th} block generates the input $x_{t,i+1}$ at the $i+1^{th}$ block through: $x_{t,i+1} = \mathcal{F}(z_{t,i})$.

Taking a sequence of video frames as the network inputs, Δ - Σ modulation leverages the distributive property of linear transformation, calculating the output z_t of convolution $\mathcal{G}$ with z_{t-1} , x_{t-1} and x_t as follow:

$$\begin{aligned} z_t &= \mathcal{G}(x_{t-1}) + \mathcal{G}(x_t) - \mathcal{G}(x_{t-1}) \\ &= \mathcal{G}(x_{t-1}) + \mathcal{G}(x_t - x_{t-1}) \\ &= z_{t-1} + \mathcal{G}(\Delta x_t), \end{aligned} \quad (1)$$

where Δx_t represents the delta activation changes between time $t-1$ and time t : $\Delta x_t = x_t - x_{t-1}$, z_{t-1} denotes the accumulation of $\mathcal{G}$ outputs at time $t-1$, and z_t is the updated output accumulation at time t with the new delta input Δx_t . Here, z is regarded as the neuron state for convolution $\mathcal{G}$, any compression on z can disrupt the linearity of Eq. (1).

To further enhance the temporal sparsity, MEET employs

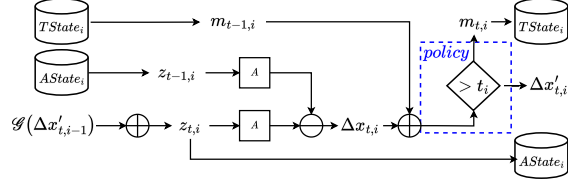


Figure 5. Temporal Δ - Σ modulation [18] on the i^{th} convolution $\mathcal{G}$ with delta event truncation at time t . A_{state} is the accumulation state while T_{state} is the truncation state.

EvNet [18] as the event firing mechanism due to its superior performance in event suppression without compromising accuracy, as shown in Fig. 5. Besides, MEET uses power-of-2 layer-specific thresholds for hardware efficiency.

4.2. Guidelines for Activation $\rightarrow$ Weight Tradeoff

Guideline 1: Fusing layers with low static computations for activation reduction Modern compute-efficient DNNs [6, 28, 51, 54, 55] are often built on the inverted bottleneck block (MB) [51], where most compute and weight savings come from depthwise convolution. However, depthwise convolution contributes negligibly to dynamic computation reduction in temporal suppression, because of its low static MACs. Inspired by previous work [25, 56], we perform layer fusion to reduce the activation count. As shown in Fig. 6, Fuse MB fuses depthwise and expansion convolutions; however, the projection convolution produces fully dense inputs for the subsequent expansion layer, resulting in no computational savings. To induce sparsity, one solution is to introduce ReLU after linear projection, termed FuseV1 MB. Another approach is to fuse depthwise and projection convolutions, where the increased static MACs can be largely offset by ReLU sparsity, referred to as FuseV2 MB. Moreover, Sandler et al. [51] show that the removal of ReLU in low-dimensional input spaces is crucial for preserving compact feature representations. Thus, FuseV2 MB is adopted as the basic convolution block in MEET.

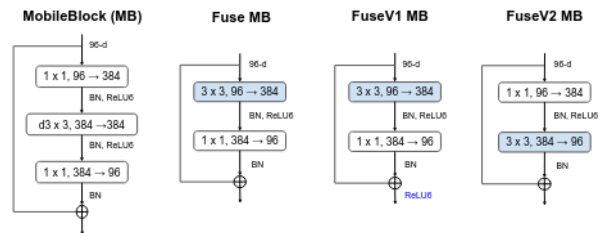
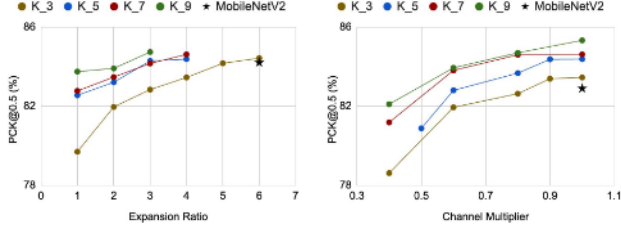


Figure 6. Convolution block design for temporal suppression.

Guideline 2: Increasing kernel size improves accuracy, allowing for a reduction in network width. Previous studies [30, 55] show that model accuracy typically scales with MACs within the same network architecture. As each MAC operation represents the inner product of an activation and a weight, we empirically observe that increasing the weight count C_w can decrease the activation count C_a without compromising accuracy, and vice versa. Furthermore, recent work show a promising trend in increasing kernel size in

DNNs to significantly boost the model accuracy by integrating reparameterization [17, 42], and weight sparsity tricks [40]. Consequently, as the kernel size increases, the network width (i.e., expansion ratio α and channel multiplier δ) can be significantly reduced at the same accuracy level, as shown in Fig. 7a and Fig. 7b. The reduction in network width can be interpreted as the elimination of feature maps, leading to a substantial decrease in activation count.



(a) Kernel size vs. Expansion Ratio (b) Kernel size vs. Ch Multiplier

Figure 7. Relation between kernel size and network width in MobileNetV2 [51] with FuseV2 MB for Body Pose Estimation [61].

Guideline 3: More efficient $A \rightarrow W$ tradeoff in deep network stages. We replace MB in MobileNetV2 with FuseV2 MB at different network stages: FuseV2-Full, FuseV2-Mix@ $8\times$ and FuseV2-Mix@ $16\times$. As shown in Fig. 8d, networks with deep stage conversion, FuseV2-Mix@ $\lambda\times$, reduce network width more than FuseV2-Full when increasing weight count. This observation is consistent with findings from other optimization studies [9, 43, 58], attributable to the increased weight redundancy in deeper stages and the reduced sensitivity to errors in high-level features.

Guideline 4: Neural Architecture Search enables more fine-grained $A \rightarrow W$ tradeoff. $A \rightarrow W$ tradeoff varies at different parts of network and can also differ depending on the dataset and application. A more detailed, fine-grained search is expected to yield a superior tradeoff. To achieve this, we employ Neural Architecture Search (NAS) to automate the process of architecture engineering for optimizing the layerwise $A \rightarrow W$ tradeoff. Drawing inspiration from prior advancements in multi-objective NAS [10, 44, 52], we employ the evolutionary search paradigm with a NSGA-III algorithm [16] to concurrently optimize three critical design objectives: accuracy, computation, and state memory footprint. In accordance with Guideline 1 and Guideline 2, we meticulously construct our search space with 8 candidates blocks, as shown in Tab. 1. Specifically, $\hat{k}$ indicates the kernel size in the initial network architecture and larger kernel size is paired with a smaller expansion ratio to optimize the $A \rightarrow W$ tradeoff. Zero Skip removes the entire block, promoting activation reduction by decreasing network depth. Furthermore, weight sharing [24] is leveraged to optimize the search for the channel number, serving as another crucial factor in minimizing activations.

A supernet is constructed using the proposed search space, adhering to the macro-architecture of EfficientNet [55], to

Table 1. Configurations of candidate blocks in the search space.

Block Type	Expansion	Kernel
Fuse V2 MB	4	$\hat{k}$
Fuse V2 MB	6	$\hat{k}$
Fuse V2 MB	1	$\hat{k}+2$
Fuse V2 MB	3	$\hat{k}+2$
Fuse V2 MB	5	$\hat{k}+2$
Fuse V2 MB	1	$\hat{k}+4$
Fuse V2 MB	3	$\hat{k}+4$
Zero Skip	-	-

accommodate varying accuracy requirements. We perform a uniform-sampling training for supernet to optimize all the architecture weights simultaneously, formulated as:

$$\mathcal{W}_{\mathcal{A}_s} = \arg \min_{\mathcal{W}} \mathcal{L}_{\text{train}}(\mathcal{N}(\mathcal{A}_s, \mathcal{W})), \quad (2)$$

where $\mathcal{N}$ denotes the supernet, $\mathcal{A}_s$ signifies the search space, and $\mathcal{W}$ represents the weights of the entire supernet.

In the searching phase, MEET adopts Nondominated Sorting Genetic Algorithm III (NSGA-III) [15, 35] as a searching algorithm for many objective optimization problem. The searched architecture is performed as:

$$\alpha^* = \arg \min_{\alpha \in \mathcal{A}_s} (\mathcal{L}_{\text{task}}(\mathcal{N}_\alpha), \mathcal{M}_{\text{state}}(\mathcal{N}_\alpha), \mathcal{C}_{\text{cycle}}(\mathcal{N}_\alpha)), \quad (3)$$

where $\mathcal{N}_\alpha = \mathcal{N}(\alpha, \mathcal{W}_{\mathcal{A}_s}(\alpha)),$

where α represents different architectures, $\mathcal{N}_\alpha$ is the sub-network of architecture α from supernet. $\mathcal{L}_{\text{task}}(\mathcal{N}_\alpha)$ indicates the evaluation task loss for $\mathcal{N}_\alpha$, $\mathcal{M}_{\text{state}}(\mathcal{N}_\alpha)$ represents the state memory usage of temporal $\mathcal{N}_\alpha$, and $\mathcal{C}_{\text{cycle}}(\mathcal{N}_\alpha)$ stands for the static cycle count of $\mathcal{N}_\alpha$ inference. Both computation and state memory usage are considered during the search process, termed Compute- and State Memory-aware NAS (CSM-NAS). This is due to the fact that compressing neuron states often introduces redundant computations, while overly minimizing computations can lead to an increase in state memory size (more details in Appendix D).

4.3. Network and Suppression Co-Design

Previous studies [18, 69] reveal that early network stages dominate state memory usage due to their large feature sizes. Spatial suppression [37, 72], coupled with depth-first processing [3, 23, 62], induces activation sparsity and consumes negligible states. This approach can serve as an alternative to temporal suppression, helping to alleviate the state memory bottleneck. As shown in Fig. 8a, Lite2 [55] outperforms CSM-NAS in dynamic cycles across stages under spatial suppression, indicating that mobile architectures are better suited for cycle reduction with spatial suppression. Conversely, Fig. 8b shows that CSM-NAS matches Lite2 in dynamic cycles with temporal suppression. Considering that in deep network stages 1) feature size diminishes, 2) computation intensifies, and 3) $A \rightarrow W$ tradeoff improves, MEET-Mix@ $\lambda\times$ employs a mobile architecture with spatial suppression in the initial network stages, while incorporating CSM-NAS with temporal suppression in the deeper stages. This mixed approach optimizes suppression

performance while minimizing the state size. As depicted in Fig. 8, MEET-Mix@16x achieves dynamic cycle performance on par with Lite2 across all stages.

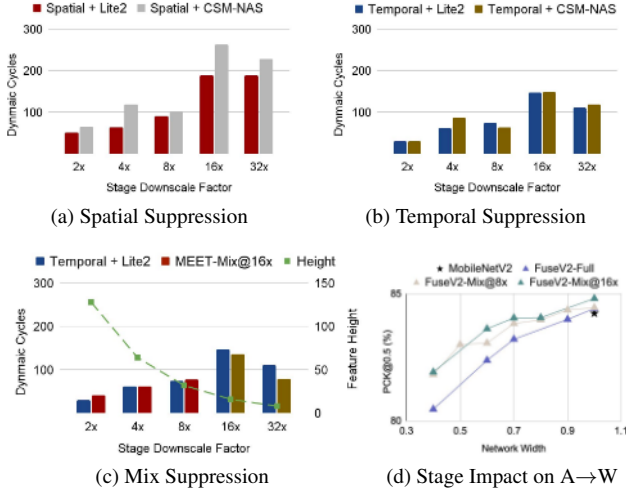


Figure 8. Stagewise analysis of network architectures and suppression methods, using SimpleBaselines [61] on MPII videos [2].

5. Experiments

5.1. Experimental Setup

Datasets and Applications: To verify the effectiveness of MEET, we conduct experiments on three vision applications: human body pose estimation on MPII [2], object detection on UA-DETRAC [60] and semantic segmentation on Cityscapes [11]. These datasets provide video sequences that are well-suited to our objective of temporal evaluation.

Network Architectures: We evaluate our approach using SimpleBaselines [61] for pose estimation, SSDLite [41] for object detection, and DeepLabV3+ [8] for semantic segmentation. We replace the backbones with widely-used network architectures (MobileNet [32], MobileNetV2 [51], ResNet [30], GhostNet [28], EfficientNet-Lite [25]) as well as CSM-NAS networks in MEET to demonstrate the effectiveness of $A \rightarrow W$ tradeoff in achieving memory savings.

Compared SOTA Methods: We establish state-of-the-art (SOTA) benchmarks by applying spatial and temporal suppression to compute-efficient mobile networks [51, 55]. As these networks are recognized as SOTA in optimizing accuracy and static MAC, integrating activation suppression can achieve SOTA dynamic computations. The compared methods fall into three categories: (1) **Memory-efficient spatial suppression:** Methods like L_1 [21], Square-Hoyer [37], STAR [72], focus on suppressing events and computations in the spatial domain. (2) **Compute-efficient temporal suppression:** Methods like RRM [49], EvNet [18], and CATS [70] maximize event suppression in temporal with less focus on state size. (3) **Memory-efficient temporal suppression:** Methods prioritize memory efficiency by skipping or compressing states (e.g., TempDiff [14], State Com-

pression (our)) or by employing mixed execution across spatial and temporal domains (e.g., DAL [64], ELSE [69]).

Evaluation Protocol: We evaluate MEET using key metrics: accuracy, event-driven dynamic cycles (Dyn Cyc.), state memory usage (State Mem.), and memory usage of weights and states (Total Mem.). **The best values for these metrics are underlined in all tables.** When comparing MEET to other suppression methods, we report the active energy for (batch=1) DNN inference, estimated on an event-driven platform [46, 59], highlighting the importance of fitting temporal models within on-chip SRAM.

5.2. Evaluation on Human Body Pose Estimation

Effect of sparsity for dynamic computations: Fig. 9 shows that FuseV2 MB initially exhibits up to $2.8\times$ static cycles than MB, but this gap narrows to $1.6\times$ with ReLU sparsity and is fully closed with temporal sparsity. This suggests that temporal sparsity is more data-dependent than architecture-dependent, allowing network redesign within hardware constraints despite increasing static computations. Notably, FuseV2 MB maintains accuracy parity with MB across different network scales, underscoring its scalability.

Effect of $A \rightarrow W$ tradeoff: Tab. 2 portrays the effectiveness of $A \rightarrow W$ tradeoff through FuseV2 MB alongside its CSM-NAS variant under MobileNetV2 [51] macro architecture. Compared to MobileBlock, FuseV2 MB reduces activations from 9.4M to 4.5M ($2.1\times$), with a further reduction to 2.7M ($3.5\times$) via CSM-NAS. Although $A \rightarrow W$ CSM-NAS initially results in increased static cycles and weight count, they are efficiently reduced to MobileBlock levels through temporal suppression [18] and weight pruning [68]. Note that the result of C_w can be further improved with the advanced weight compression techniques [57, 63].

Table 2. Effectiveness of $A \rightarrow W$ tradeoff in minimizing activation count while preserving accuracy, weight count and dynamic cycles. We evaluate models [61] on 20-frame videos [2].

Block Type	PCK @0.5 (static)	PCK @0.5 (dyn)	Static Cyc. (M)	Dyn Cyc. (M)	C_w (M)	C_a (M)
MobileBlock	84.21%	83.90%	75	24	2.3	9.4
FuseV2 MB	84.28%	84.07%	196	24	9.8	4.5
$A \rightarrow W$ (CSM-NAS)	84.47%	84.02%	223	25	13.6	2.7
+ Prune [68] 80% W	84.40%	84.02%	223	25	2.7	2.7

Comparisons with widely-used network architectures (grouped by accuracy): As shown in Tab. 3, one prominent trend is that MEET-Full (S, M, L) consistently achieves similar or lower dynamic cycles than mobile networks (MobileNetV2, ENet-Lite2, ENet-Lite4) across various accuracy levels in temporal suppression, while consuming the lowest state and total memory usage among fully temporal-executed models. For example, compared to MobileNetV2, MEET-Full S retains similar dynamic cycles, but reduces state memory from 37.4MB to 10.8MB and total mem-

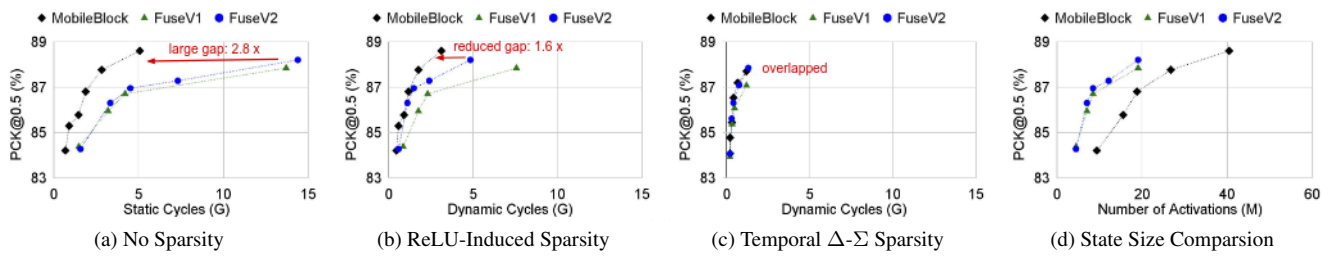


Figure 9. Comparison of cycles and activation count between MobileBlock (MB) and our blocks (FuseV1, FuseV2). We evaluate all the results using SimpleBaselines [61] with different backbones using 20-frame 256×256 MPII [2] videos. FuseV2 consistently achieves much fewer activations than MB, while maintaining dynamic cycles through temporal suppression [18] across various accuracy levels.

Table 3. **Comparisons with other network architectures.** Different backbones are evaluated on 20-frame MPII [2] videos. The best value is in bold and blue, and the second-best value is in bold.

Backbone	PCK @0.5	Static Cyc. (M)	Dyn Cyc. (M)	Param Mem. (MB)	State Mem. (MB)	Total Mem. (MB)
Input Resolution: 256×256						
MobileNetV1 [32]	83.25%	113	27	1.6	26.9	28.5
MobileNetV2 [51]	83.90%	75	24	1.2	37.4	38.6
VGGBN16 [53]	83.54%	777	59	4.5	24.5	29.0
ResNet18 [30]	83.85%	296	29	2.8	11.4	14.2
MEET-Full S	84.02%	233	25	3.4	10.8	14.2
MEET-Mix@16×S	83.80%	184	25	2.7	2.8	5.5
Input Resolution: 296×296						
ENet-Lite2 [55]	86.40%	218	53	2.4	66.2	68.6
MEET-Full M	86.39%	742	53	6.7	26.6	33.2
MEET-Mix@16×M	86.47%	561	54	6.0	6.8	12.8
Input Resolution: 342×342						
ENet-Lite4 [55]	87.86%	614	151	5.8	151.0	156.9
MEET-Full L	87.72%	1798	145	13.7	76.3	90.0
MEET-Mix@16×L	87.78%	1741	144	14.9	16.9	31.8

ory from 38.6MB to 14.2MB—a $3.5\times$ and $2.7\times$ decrease, respectively. A second notable trend is that MEET-Full S achieves lower dynamic cycles and state memory than ResNet18 and VGGBN16, indicating that temporal sparsity can substantially narrow the computation gap, but can not entirely equalize it across all networks. This underscores the efficacy of CSM-NAS in balancing computation and state size in the $A \rightarrow W$ tradeoff. Finally, MEET can further optimize memory usage by shifting from full-temporal to mixed execution, Mix@16× (S, M, L) significantly reduces state and total memory usage compared to temporal mobile networks while preserving accuracy and dynamic cycles. This highlights the effectiveness of MEET in reducing memory demands for temporal $\Delta\text{-}\Sigma$ DNNs.

Comparisons with SOTA suppression methods (grouped by accuracy): We employ EfficientNet-Lite2 [55] as the backbone for SimpleBaselines [61], with its ReLU-sparsified variant serving as the baseline for performance comparison. Tab. 4 shows that temporal methods outperform spatial methods in cycle reduction but exceed the on-chip memory constraints of the evaluated event-driven platform. Consequently, the expensive off-chip access leads to $4.7\sim 7.8\times$ more energy-consuming than running the baseline entirely within SRAM, negating the benefits of computation savings. The memory-efficient method TempDiff[‡]

Table 4. **Comparisons with SOTA methods on memory usage and hardware performance.** The models are evaluated with 20-frame 256×256 MPII [2] videos. Hardware cycle and normalized energy E'_{active} are estimated for an event-driven platform GrAI-VIP, equipped with 36 MB on-chip SRAM [46, 59]. The ReLU-sparsified variant is considered the baseline. †, ‡, and ◇ represent spatial, temporal, and mixed execution, respectively.

Method	PCK @0.5	Dyn Cyc. (M)	State Mem. (MB)	Total Mem. (MB)	SRAM only	E'_{active}
Baseline (ReLU [†])	86.81%	137 (1.0×)	0	4.8	✓	1.00 (1.0×)
Hoyer [†] [37]	86.49% (-0.37%)	108 (1.3×)	0	4.8	✓	0.81 (1.2×)
STAR [†] [72]	86.43% (-0.44%)	83 (1.7×)	0	4.8	✓	0.64 (1.6×)
EvNet [‡] [18]	86.40% (-0.47%)	53 (2.6×)	66.2	70.0	✗	7.40 (7.4×)
CATS [‡] [70]	86.39% (-0.48%)	68 (2.0×)	33.1	37.9	✗	4.69 (4.7×)
TempDiff [‡] [14] (reset=2)	83.68% (-3.61%)	75 (1.8×)	27.5	32.2	✓	0.55 (1.8×)
RRM [‡] [49] (reset=5)	85.86% (-1.09%)	67 (2.0×)	33.1	37.9	✗	7.80 (7.8×)
MEET-M [‡]	86.39% (-0.48%)	53 (2.6×)	26.6	33.2	✓	0.36 (2.8×)
ELSE [◇] [69]	86.38% (-0.50%)	77 (1.8×)	11.2	16.0	✓	0.60 (1.7×)
MEET-M [◇]	86.47% (-0.39%)	54 (2.5×)	6.8	12.8	✓	0.37 (2.7×)

provides the lowest total memory usage but incurs an accuracy drop of 3.6%. MEET-M[‡] attains dynamic cycles on par with EvNet, the most cycle efficient among all methods, while meeting SRAM constraints. It efficiently converts cycle reduction into energy savings, exhibiting reductions of $2.6\times$ in cycles and $2.8\times$ in energy consumption, relative to the baseline. Moreover, ELSE[◇] also fits within SRAM through mixed execution. However, MEET-M[◇] achieves the lowest state and overall memory usage while significantly outperforming ELSE in cycle and energy savings. This highlights the effectiveness of our co-design of network architectures and suppression methods.

Comparisons with SOTA suppression methods in long-term video processing without keyframe correction (grouped by dynamic cycles): A key highlight of temporal suppression is that $\Delta\text{-}\Sigma$ modulation leverages temporal redundancy in video while preserving accuracy in long-term

processing. As illustrated in Fig. 10, EvNet [18] maintains accuracy in long-term processing with heavy states. Tempdiff [14] skips nonlinear layers for intermediate state removal, resulting in a $\sim 4\%$ accuracy drop with only 2 frame processing. RRM [49] consumes half of EvNet states; however, the lack of long-term memory results in accuracy degradation after just a few frames. STAR [72] does not require states for temporal dependency, but results in over 1% accuracy drop at a similar computation level. State compression [67], an approach that we initially developed, maintains a relatively small accumulated error but still requires frequent keyframe corrections. Finally, both MEET-Full and MEET-Mix fully prevent accuracy degradation in long-term processing, making them optimal candidates.

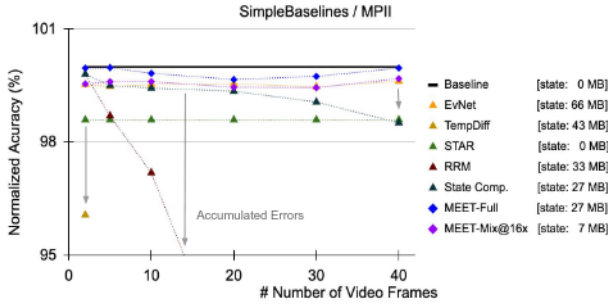


Figure 10. Comparisons with SOTA event suppression methods in long-term video processing without keyframe correction.

5.3. Evaluation on Other Applications

To demonstrate the broader applicability of MEET, we evaluate its performance on the surveillance dataset UA-DETRAC for object detection and the car driving dataset Cityscapes for semantic segmentation.

Object Detection: We evaluate various network backbones for SSDLite [41]. Tab. 5 shows that MEET-Full outperforms other networks in all key metrics. Compared to MobileNetV2, it attains a higher mAP@0.5, rising from 50.15% to 51.50%, while reducing dynamic cycles from 7M to 5M and lowering total memory usage from 86.4 MB to 26.7 MB. Notably, MEET-Mix@ $\lambda\times$ variants lag behind MEET-Full in dynamic cycle efficiency, a gap that widens with a reduction in temporal-executed layers. We attribute this to the high temporal redundancy in surveillance videos, resulting in significant temporal sparsity that spatial suppression fails to offset. This implies that λ should be adjusted based on the deployment scenarios (see Appendix E).

Effect of $A \rightarrow W$ tradeoff: Tab. 6 demonstrates that, under the same SRAM constraints, $A \rightarrow W$ tradeoff yields three distinct advantages. First, it allows more layers to run temporally, reducing dynamic cycles from 11M to 5M. Second, it facilitates the temporal execution of large-scale models, enhancing mAP@0.5 from 50.90% to 52.61%. Lastly, it supports higher input resolution (384×384 to 512×512), further elevating mAP@0.5 from 50.90% to 64.39%.

Semantic Segmentation: We optimize the MobileNetV2

backbone in DeepLabV3+ [8] using MEET. Tab. 7 shows that MEET-Mix@ $16\times$ improves mIoU from 50.24% to 50.93%, while reducing dynamic cycles from 64M to 60M and lowering total memory usage from 87 MB to 17 MB, compared to MobileNetV2. Surprisingly, it also outperforms MEET-Full in dynamic cycles (66M vs. 60M), which contrasts with findings in object detection. We deduce that significant camera motion in car driving limits the effectiveness of temporal suppression in the early stages, while spatial methods can maintain or surpass suppression effect.

Table 5. **Object Detection.** The models are evaluated on 30-frame 384×384 video frames. The best value in each column is highlighted in bold and blue, and the second-best value is in bold.

Backbone	mAP @0.5	Static Cyc. (M)	Dyn Cyc. (M)	Param Mem. (MB)	State Mem. (MB)	Total Mem. (MB)
ResNet18 [30]	51.13%	666	10	2.8	25.6	28.4
GhostNet130 [28]	48.94%	140	6	1.9	78.8	80.7
MobileNetV1 [32]	48.62%	253	6	3.2	60.4	63.6
MobileNetV2 [51]	50.15%	168	7	2.2	84.2	86.4
MEET-Full	51.50%	133	5	2.0	24.7	26.7
MEET-Mix@ $8\times$	51.60%	90	8	0.7	5.3	6.0
MEET-Mix@ $16\times$	50.50%	106	13	0.8	2.8	3.6

Table 6. **Object Detection.** Effectiveness of $A \rightarrow W$ tradeoff.

Mix Variants	Input Resol.	mAP @0.5	Dyn Cyc. (M)	State Mem. (MB)	SRAM only
MNV2-Mix@ $16\times$	384×384	50.90%	11	21	✓
More Temporal Execution					
MNV2-Full	384×384	50.15%	7	84	✗
MEET-Full [‡]	384×384	51.50%	5	27	✓
Larger Input Resolution					
MNV2-Mix@ $16\times$	512×512	64.40%	24	38	✗
MEET-Mix@ $4\times$	512×512	64.39%	16	28	✓
Deeper Neural Network					
Lite2-Mix@ $16\times$	384×384	52.16%	20	35	✗
MEET(Lite2)-Mix@ $16\times$	384×384	52.61%	19	5	✓

Table 7. **Semantic Segmentation.** The models are evaluated on 30-frame 256×512 video frames. The best value is in bold.

Backbone	mIoU	Static Cyc. (M)	Dyn Cyc. (M)	Param Mem. (MB)	State Mem. (MB)	Total Mem. (MB)
MobileNetV2 [51]	50.24%	234	64	1.8	85.2	87.0
MEET-Full	49.66%	531	66	2.7	42.6	45.4
MEET-Mix@ $16\times$	50.93%	1299	60	6.0	11.0	17.0

6. Conclusion

Temporal $\Delta-\Sigma$ DNNs excel in computation savings for video processing but often exceed the on-chip memory limits of embedded platforms. MEET addresses this challenge by reducing the activation size and applying a mixed spatial-temporal execution. It achieves substantial memory savings without compromising accuracy and dynamic computations in long-term $\Delta-\Sigma$ processing. This enables on-chip deployment of temporal $\Delta-\Sigma$ DNNs, leveraging near-memory and sparsity-aware computing for low-power edge processing.

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